

Senspark: Inventing a limitless future for blockchain games with Google Cloud

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ABOUT SENSPARK

Through blockchain, Senspark has built BombCrypto, Seraph, a community of over 1.2 million players, and an agile tokenomics business model, all of which have devised a strategy for unlimited gaming powered by Google Cloud.



About Senspark

Senspark is a Vietnamese gaming startup on a mission to enable participation in the decentralized global digital economy. It offers a game-based metaverse with entertainment and finance. Since launching in 09/2021, Senspark's blockchain game, BombCrypto, has garnered more than 2 million registered users and minted over 20 million NFTs, with a daily NFT trading volume of more than \$2 million. Senspark aims to take a leading role in the GameFi market that empowers gamers financially while having fun.

Google Cloud results

- Supports rapid growth to 2 million registered gamers with zero service interruptions through Kubernetes autoscaling
- Frees up funds for game development by saving 80% in infrastructure budget through GKE migration
- Enables continuously updated game tokenomics by processing up to 1 billion live player events per

Installed to 600 concurrent online

Industries: Media & Entertainment

Location: Vietnam

SQL (<https://cloud.google.com/sql/>),



About Cloud Kinetics

Cloud Kinetics is a leading Cloud Transformation Solutions and Managed Services Provider, specializing in the design, build, deployment, migration and operation of complex cloud environments for enterprise clients across industries. A portfolio company of I Squared Capital, the firm has helped hundreds of customers unlock new-found value by running business critical systems in the cloud.

Google Cloud (<https://cloud.google.com/>)

Transformation Solutions and Managed
BigQuery (<https://cloud.google.com/bigquery/>)

Cloud SQL (<https://cloud.google.com/sql/>) and

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)
portfolio company of I Squared Capital, the
App Engine (<https://cloud.google.com/appengine/>)

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day through BigQuery

For more than a decade, Vietnamese gamemaker [Senspark](https://senspark.com/) (<https://senspark.com/>) has been developing hit multiplayer online games, downloaded more than 100 million times by fans across Asia and beyond.

When non-fungible tokens (NFTs) took the world by storm, driven by multimillion dollar sales of the blockchain-based digital assets, Senspark saw an opportunity to "do something massive." It decided to move its gaming experience into the metaverse, where gamers could play for NFTs worth real money in a virtual online world.

In this GameFi vision, which merges gaming with decentralized finance (DeFi), gaming "moves beyond playing for fun to playing to earn," as fans trade and battle for NFT assets, explains Senspark's Marketing Lead, Nhung Vo. Launching its NFT game BombCrypto in early 2021, Senspark saw its fan base soar beyond expectations, with the game surpassing more than 2 million registered users in one year.



The overnight success of BombCrypto also created challenges. Senspark found its on premise servers struggling to avoid crashes as fans flocked to the game, and unable to scale seamlessly amid wild swings in concurrent users. The developers also needed to build the game's tokenomics, the system of rewards and incentives that govern the value of assets in crypto transactions. That required the most powerful cloud-based data analytics tools available.

After evaluating several cloud providers, Senspark decided to "go all-in" with Google Cloud (<https://cloud.google.com/>) thanks to what it saw as the more robust, versatile and seamless autoscaling available provided by Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) and BigQuery (<https://cloud.google.com/bigquery>).

According to Hai Hoang, Lead Engineer, Senspark, the decision unlocked a gaming future of unlimited potential for growth and fandom in the Senspark metaverse.

"With the instant success of our NFT game BombCrypto, we suddenly faced the need to achieve

massive scale," says Hoang. "On Google Cloud, we can deliver an uninterrupted experience to users, even as our fan base grows incredibly fast, while driving the frontiers of gaming tokenomics."

Game-changing cloud transformation for "playing to earn" in the metaverse

Working with [Cloud Kinetics](https://www.cloud-kinetics.com/)

(<https://www.cloud-kinetics.com/>), Senspark carried out a comprehensive migration to a Google Cloud architecture at a time of explosive growth in BombCrypto popularity, without a single interruption in service. This was thanks largely to the detailed configuration guidance that Cloud Kinetics provided in the GKE setup, which enabled a smooth transition in two months.

The move to Google Cloud was transformative, says Hoang. With the seamless scalability of GKE, Senspark is able to scale instantly to more than 600,000 concurrent users in a day, and just as easily scale down to 300,000 during quiet periods. In less than a year, BombCrypto has driven a NFT trading volume of over \$640 million, with more than 20 million minted NFTs.

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*—Hai Hoang, Lead
Engineer, Senspark*

"Imagine needing to maintain computing capacity at more than 1 million users for system stability, in an on-prem environment. Multiple servers would need to be up continuously, a drain on resources when demand is low," says Hoang. "Thanks to GKE we can ride the waves of gaming spikes and troughs, automatically scaling up and down no matter how

big the swings. This keeps fans happy and saves on costs."

Hoang estimates the cost savings moving from on-prem to a GKE-based environment at roughly 80% of its total infrastructure budget, freeing up funds for game development. Moreover, adds Hoang, Senspark is able to drive rapid expansion of its gaming platform without adding to its lean DevOps team of two engineers. Hosted on its own servers, says Hoang, Senspark would need to regularly hire more engineers simply to provision servers quickly to meet spikes in demand.

"Google Cloud helps us to grow at unlimited scale and also save on costs, since we have less engineering to carry out to serve a massive number of users," says Hoang. "That also means our team can focus on the exciting side of creating games instead of mundane infrastructure."

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Pushing the boundaries of gaming tokenomics with BigQuery and Cloud Spanner

To build a successful "play to earn" game, Senspark needs to constantly evolve the tokenomics of its platform, providing finely-calibrated incentives for users to compete for NFT rewards. To do this, Senspark must monitor and analyze user behavior data in real time, including factors such as gaming achievements and trading frequency.

With this live information, Senspark's game economists can run the data stored in CloudSQL (<https://cloud.google.com/sql>) through BigQuery for granular insight into correct NFT values that will incentivize gamers to compete for them. Building the economy of the Senspark metaverse requires crunching staggering amounts of data. Hoang says BigQuery powers this vital task by processing up to 1 billion live player events per day, enabling a constantly updated panorama of the BombCrypto economy.

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—Hai Hoang, Lead Engineer, Senspark

Senspark now aims to further boost the precision of its tokenomics by adopting Cloud Spanner (<https://cloud.google.com/spanner/>) as its distributed SQL database management and storage service. By offering unlimited scale and up to 99.999% availability, Cloud Spanner promises to bring a new dimension to the speed and granularity of gamer analytics through BigQuery.

That in turn will enable the platform to offer even more targeted and precise rewards and incentives for its game, enhancing both enjoyment and value for millions of players.

"Deploying powerful scalable solutions such as Cloud Spanner, we see new avenues for fine-tuning the tokenomics of our blockchain game, and this is vital to driving player loyalty and our own business model," says Hoang. "That opens up a future of unlimited growth and evolution, as we create new experiences at the frontiers of GameFi."